CSTE4103 · CLOUD COMPUTING · PRACTICAL

Lab 1

Exploring Cloud Hosting Platforms

Portal Walkthrough · Specs & Pricing Comparison

⚠ Offline session: campus power outage (fuel shortage). No live browsing today — this deck substitutes for the hands-on portal walkthrough.

SESSION FORMAT

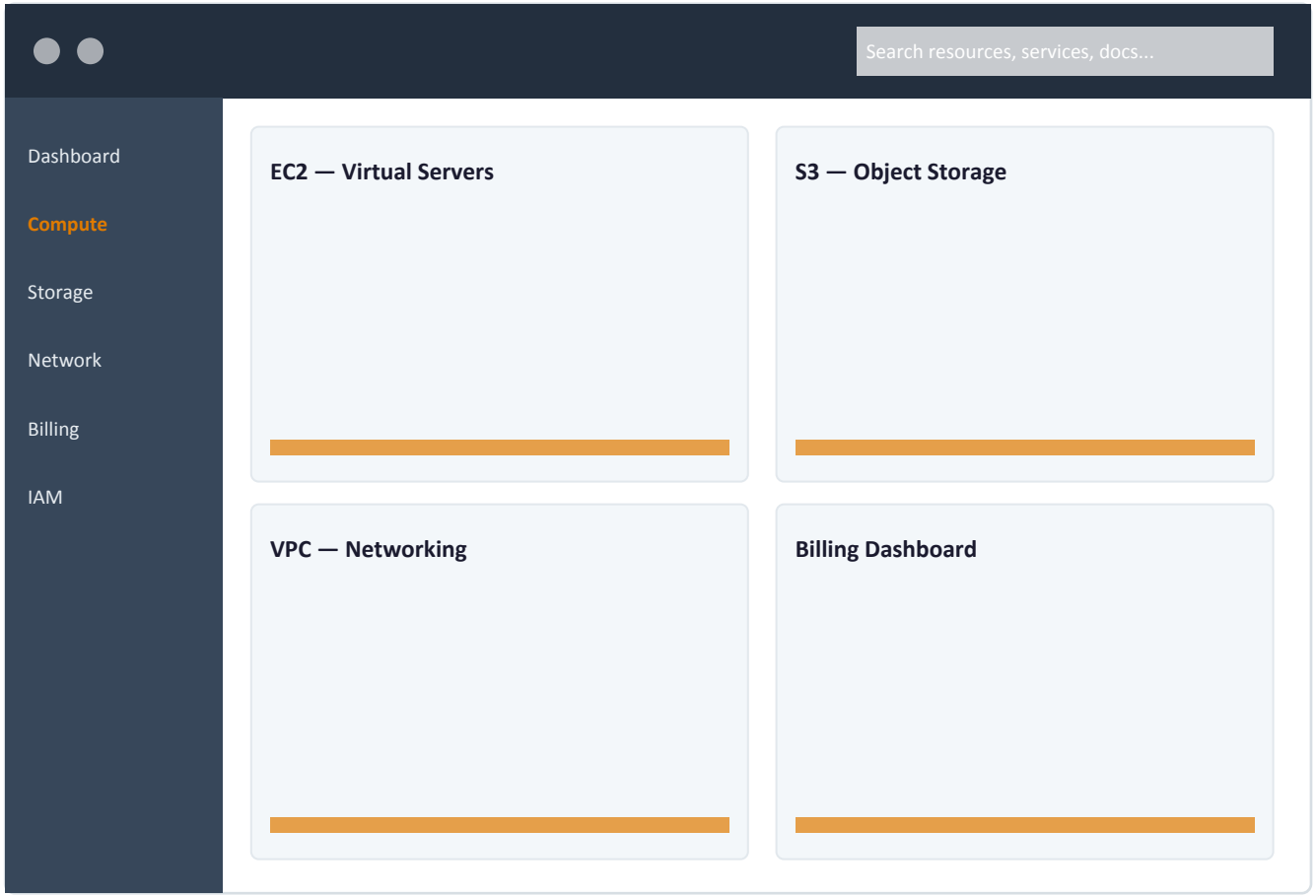
Lab Objectives

1. Recognize the layout and core sections of a major cloud provider's management console
2. Identify where compute, storage, and billing information live inside a cloud portal
3. Compare virtual-machine specs (vCPU, RAM) across AWS, Azure, GCP, and DigitalOcean
4. Compare on-demand pricing and free-tier offers across the same four providers
5. Understand the difference between on-demand, reserved/committed, and spot pricing

When power and connectivity return, this lab converts directly into a live, hands-on session — same objectives, real portals.

PORTAL 1

AWS Management Console

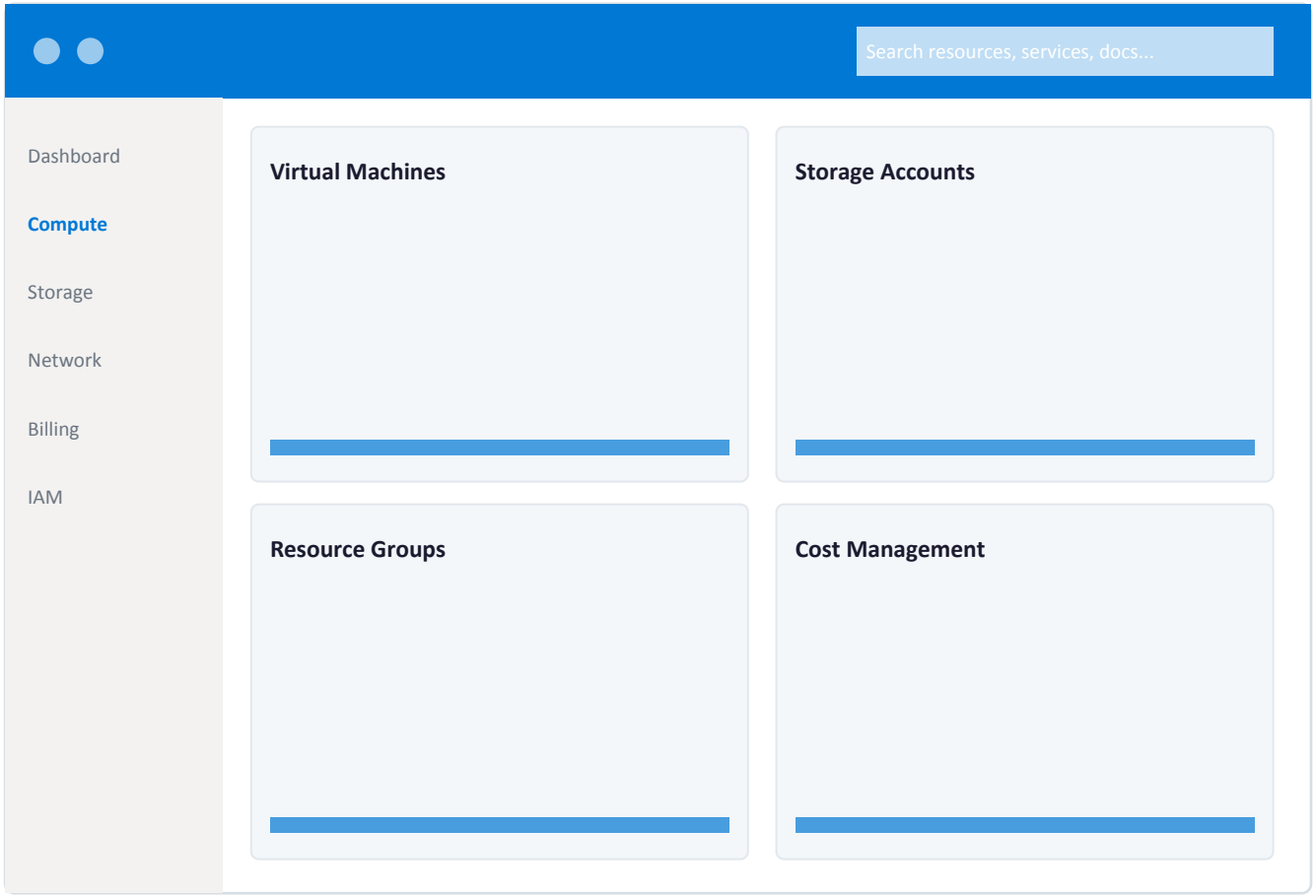


Conceptual layout — illustrative, not a live screenshot

- Region selector (top-right): every resource lives in a chosen geographic region
- Search bar: jump directly to any of 200+ AWS services by name
- Left navigation: organized by service category (Compute, Storage, Database...)
- Billing Dashboard: real-time cost tracking and Free Tier usage alerts

PORTAL 2

Microsoft Azure Portal

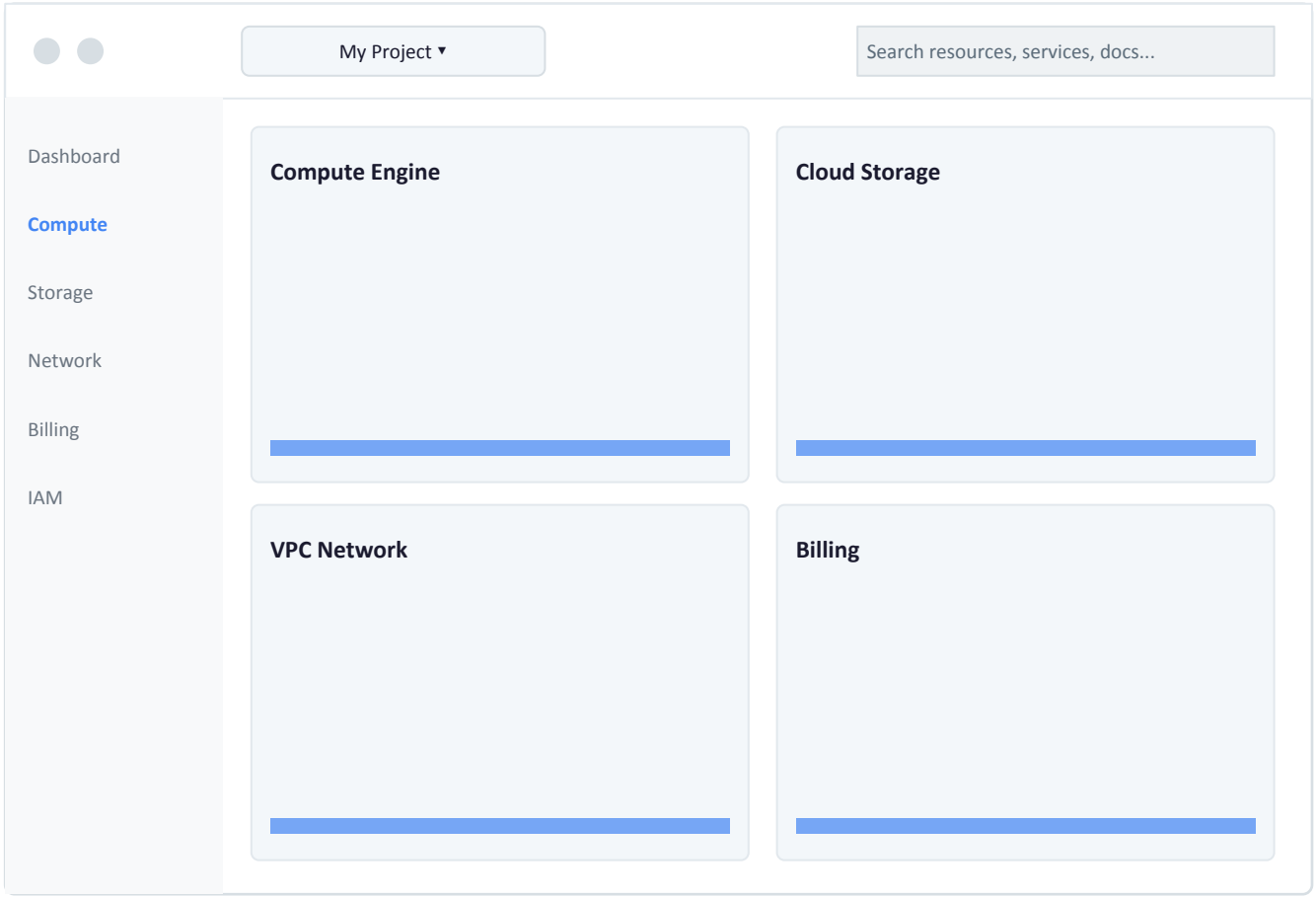


Conceptual layout — illustrative, not a live screenshot

- Resource Groups: Azure's way of bundling related resources (VM + storage + network) as one manageable unit
- Dashboard is customizable — pin any resource or metric tile you use often
- Cost Management + Billing: budgets, alerts, and free-credit tracking in one place
- Directory switcher (top bar): move between organizational tenants — relevant for enterprise/university accounts

PORTAL 3

Google Cloud Console



Conceptual layout — illustrative, not a live screenshot

- Project selector: everything in GCP belongs to a Project — the core organizing unit (comparable to an Azure Resource Group)
- Navigation menu (☰, top-left): a searchable, collapsible list of all Google Cloud products
- Cloud Shell: a free, browser-based terminal — no local setup needed for command-line work
- Billing: prominently separated from resources; GCP is known for a comparatively simple pricing calculator

CROSS-PROVIDER PATTERN

Every Console Shares the Same Anatomy

	AWS	Azure	GCP
Organizing unit	Account → Region	Subscription → Resource Group	Organization → Project
Compute service	EC2	Virtual Machines	Compute Engine
Object storage	S3	Storage Account (Blob)	Cloud Storage
Identity & access	IAM	Microsoft Entra ID	Cloud IAM
Billing view	Billing Dashboard	Cost Management	Billing

The vocabulary changes, but the shape stays the same — once you know one console, orienting in another is mostly relearning names.

Entry-Level Compute: Specifications

Provider	Entry Instance	vCPU	RAM	Best For
AWS	t3.micro	2 (burstable)	1 GiB	General-purpose dev/test workloads
Azure	B1s	1 (burstable)	1 GiB	Lightweight utility tasks, dev boxes
GCP	e2-micro	2 (shared, burstable)	1 GiB	Low-traffic web servers, small DBs
DigitalOcean	Basic Droplet (1GB)	1 (shared)	1 GB	Simple apps, learning, small sites

All four are “burstable” entry instances — normal CPU usage is capped low, with short bursts to full speed allowed. Ideal for a lab VM; not for sustained heavy load.

COMPREHENSIVE COMPARISON — PART 2

Entry-Level Compute: Pricing & Free Tier

Provider	On-Demand (approx.)	Free Tier Offer
AWS	~$0.0104/hr (~$7.59/mo)	750 hrs/mo of t2/t3.micro — 12 months
Azure	~$0.0104/hr (~$7.59/mo)	750 hrs/mo of B1s — 12 months + $200 credit (30 days)
GCP	~$0.0084/hr (~$6.11/mo)	1 e2-micro instance/mo — Always Free (select US regions) + $300 credit (90 days)
DigitalOcean	flat $6.00/mo	No perpetual free tier (promotional credit only, varies)

Prices change often and vary by region — treat these as reference figures captured for this lecture, not live quotes. Always confirm on the provider's official pricing page before making a real decision.

On-Demand vs. Reserved vs. Spot

On-Demand

Pay per hour (or second) of actual use. No commitment, highest per-hour price.

Use it for:

Short-lived labs, unpredictable workloads, testing

Reserved / Committed

Commit to 1 or 3 years upfront in exchange for a large discount (30–60% typical).

Use it for:

Steady, predictable production workloads

Spot

Bid for a provider's unused capacity at a steep discount (up to 90% off) — can be reclaimed with little notice.

Use it for:

Batch jobs, fault-tolerant or interruptible workloads

WRAP-UP

Summary & When We're Back Online

Today, in one sentence:

All major cloud consoles share the same anatomy — region/project selector, service navigation, and a billing view — and their entry-level VMs converge on roughly the same 1-2 vCPU / 1 GB starting point, priced within a dollar or two of each other.

When connectivity returns

Log into one provider's free-tier account, locate the four sections covered today (Compute, Storage, Networking, Billing), and take one screenshot of each for your lab notebook.

Reflection Question

Given today's comparison, which provider would you pick for a personal project with zero budget — and why? Be ready to defend your answer next session.